Evaluating Prediction Intervals

Study Using the M3 Competition Dataset

Igor Faluhelyi

2024-11-27

Modelos sem *bagging*

STM

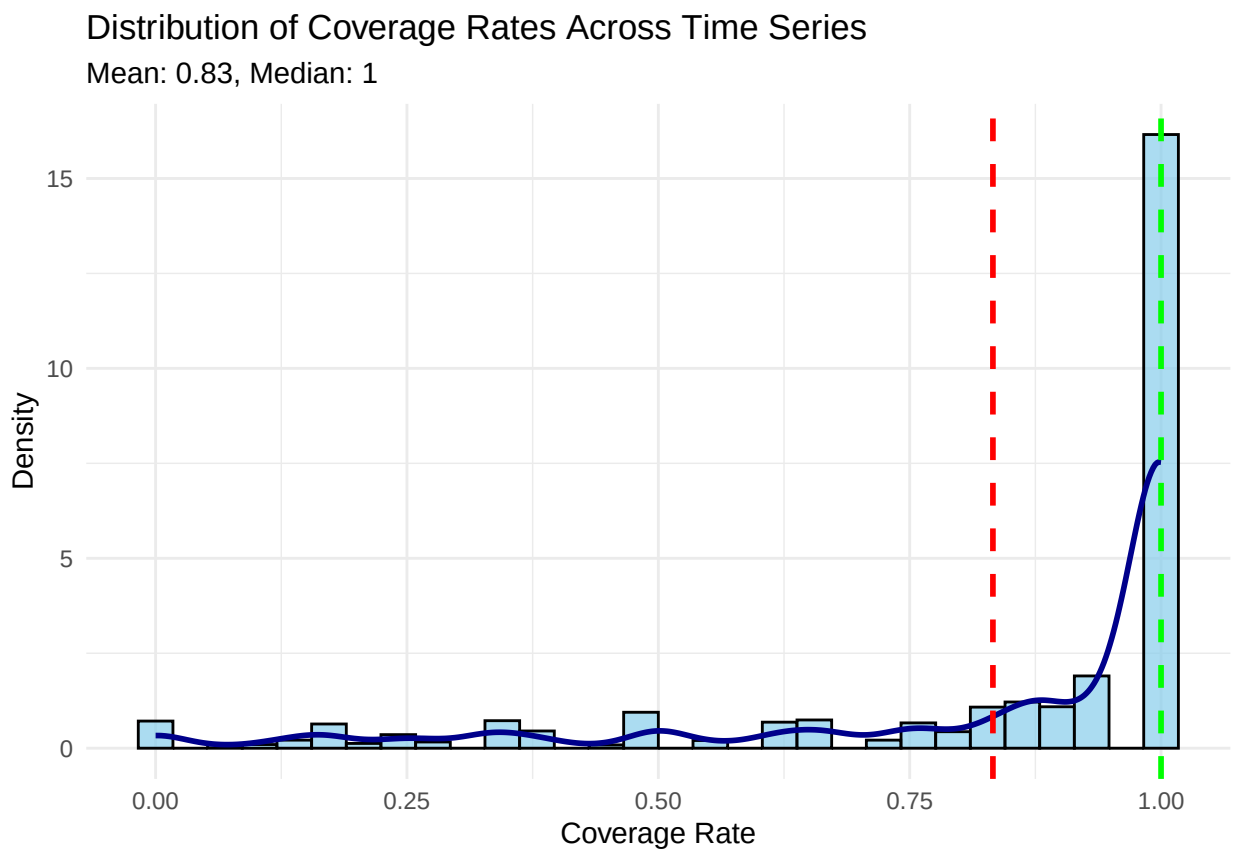
```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_stm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_stm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## i The deprecated feature was likely used in the forecTheta package.
##   Please report the issue at <https://github.com/jafiorucci/forecTheta/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
results$summary
```

```
## $mean
## [1] 0.8327737
##
## $median
## [1] 1
##
## $sd
## [1] 0.272626
##
## $min
## [1] 0
##
## $max
## [1] 1
```

OTM

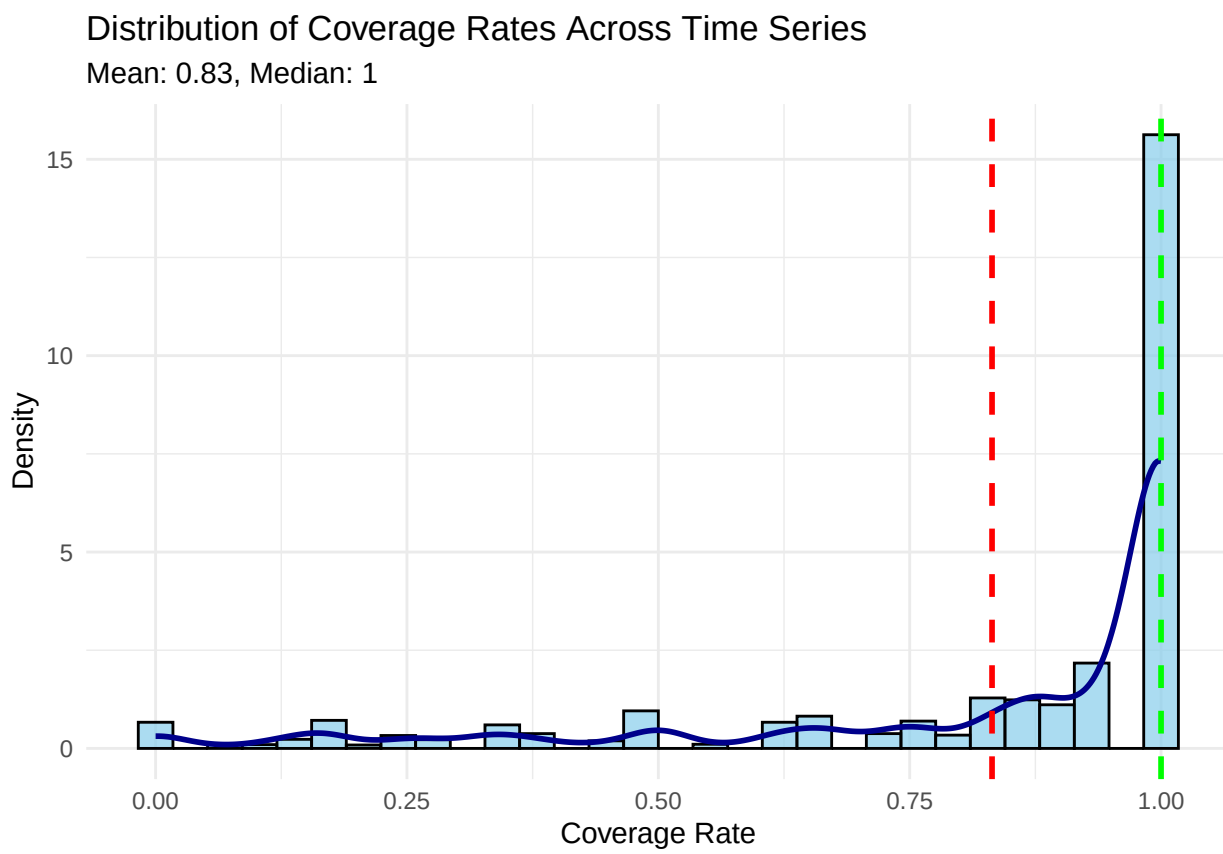
```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
```

```r
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_otm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_otm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```r
results$summary
```

```
## $mean
## [1] 0.8319135
##
## $median
## [1] 1
##
## $sd
## [1] 0.270374
##
## $min
## [1] 0
```

```
##
## $max
## [1] 1
```

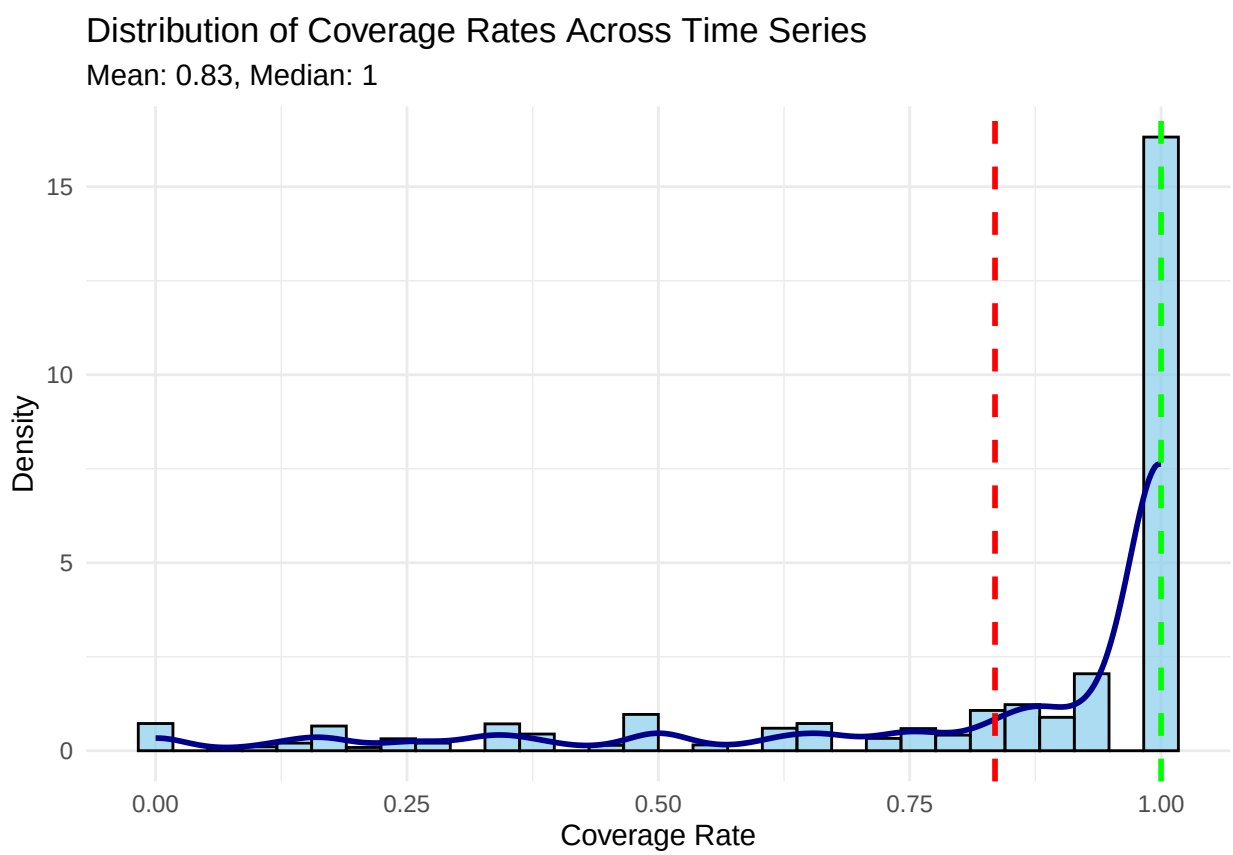
DSTM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_dstm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_dstm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```



```r
results$summary
```

```
## $mean
## [1] 0.8348873
##
## $median
## [1] 1
##
## $sd
## [1] 0.2721939
##
## $min
## [1] 0
##
## $max
## [1] 1
```

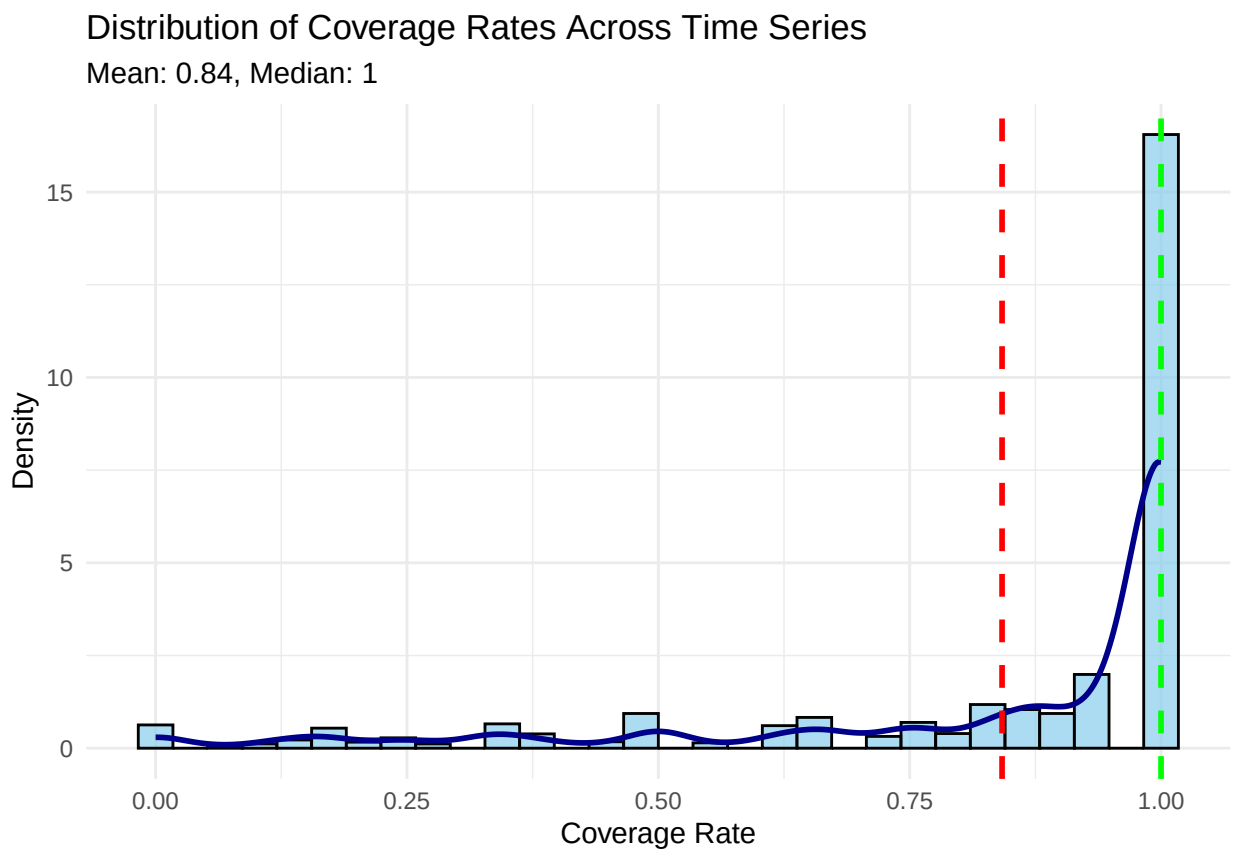
DOTM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_dotm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_dotm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```r
results$summary
```

```
## $mean
## [1] 0.8418711
##
## $median
## [1] 1
##
## $sd
## [1] 0.264369
##
## $min
## [1] 0
##
## $max
## [1] 1
```